

Ramadhana_BI_Batch_2



Query SQL

- T1_monthly_query

```
SELECT
    DATE_TRUNC('month', o.order_date) AS month,
    COUNT(DISTINCT o.order_id) AS total_orders,
    SUM(od.unit_price * od.quantity * (1 - od.discount)) AS total_revenue,
    AVG(od.unit_price * od.quantity * (1 - od.discount)) AS avg_order_value
FROM orders o
JOIN order_details od
    ON o.order_id = od.order_id
WHERE EXTRACT(YEAR FROM o.order_date) = 2025
GROUP BY DATE_TRUNC('month', o.order_date)
ORDER BY month;
```

- T2_sla_Query

```
SELECT
  s.company_name AS shipper_name,
  COUNT(o.order_id) AS shipped_orders,
  SUM(CASE WHEN o.shipped_date > o.required_date THEN 1 ELSE 0 END) AS late_orders,
  ROUND(
    SUM(CASE WHEN o.shipped_date > o.required_date THEN 1 ELSE 0 END)::numeric
    / COUNT(o.order_id) * 100,
    2
  ) AS late_rate_percent
FROM orders o
JOIN shippers s
  ON o.ship_via = s.shipper_id
WHERE o.shipped_date IS NOT NULL
  AND EXTRACT(YEAR FROM o.shipped_date) = 2025
GROUP BY s.company_name
ORDER BY late_rate_percent DESC;
```

- T3_top_products_query

```
SELECT
    products.product_name,
    categories.category_name,
    products.unit_price,
    CASE
        WHEN products.unit_price < 10 THEN '<10'
        WHEN products.unit_price BETWEEN 10 AND 20 THEN '10-20'
        WHEN products.unit_price BETWEEN 20 AND 50 THEN '20-50'
        ELSE '>50'
    END AS price_range,
    ROUND(SUM(order_details.unit_price * order_details.quantity * (1 - order_details.discount))::numeric, 2) AS
total_revenue
FROM order_details
JOIN orders ON order_details.order_id = orders.order_id
JOIN products ON order_details.product_id = products.product_id
JOIN categories ON products.category_id = categories.category_id
WHERE EXTRACT(YEAR FROM orders.order_date) = 2025
GROUP BY products.product_name, categories.category_name, products.unit_price
ORDER BY total_revenue DESC
LIMIT 10;
```

- T4_sales_rep_query

```
SELECT
    employees.first_name || ' ' || employees.last_name AS employee_name,
    COUNT(order_summary.order_id) AS total_orders,
    ROUND(AVG(order_summary.order_total)::numeric, 2) AS avg_order_value,
    ROUND(
        (SUM(CASE WHEN order_summary.order_total > 1000 THEN 1 ELSE 0 END)::numeric / COUNT(order_summary.order_id)) *
100,
        2
    ) AS pct_large_orders
FROM (
    SELECT
        orders.order_id,
        orders.employee_id,
        SUM(order_details.unit_price * order_details.quantity * (1 - order_details.discount)) AS order_total
    FROM orders
    JOIN order_details ON orders.order_id = order_details.order_id
    WHERE EXTRACT(YEAR FROM orders.order_date) = 2025
    GROUP BY orders.order_id, orders.employee_id
) AS order_summary
JOIN employees ON order_summary.employee_id = employees.employee_id
GROUP BY employees.first_name, employees.last_name
ORDER BY avg_order_value DESC;
```

- T5_segments_query

```
SELECT
  CASE
    WHEN EXTRACT(YEAR FROM (
      SELECT MIN(order_date)
      FROM orders AS o2
      WHERE o2.customer_id = o1.customer_id
    )) = 2025 THEN 'NEW'
    ELSE 'LOYAL'
  END AS segment,
  COUNT(o1.order_id) AS orders,
  ROUND(SUM(od.unit_price * od.quantity * (1 - od.discount))::numeric, 2) AS revenue
FROM orders AS o1
JOIN order_details AS od
  ON o1.order_id = od.order_id
WHERE EXTRACT(YEAR FROM o1.order_date) = 2025
GROUP BY segment;
```